Exact Convex Coding and Hyperbolic Ledgers for the Equilateral Three-Disk Open Billiard

Route-A source-local certificate HCS-C294

2 September 2026

Abstract

Three equal circular obstacles of radius r have centers at an equilateral triangle of side d. In the sharp strict no-eclipse chamber $d > 4r/\sqrt{3}$, we prove that cyclically reduced cyclic classes are in bijection with periodic-ray iterates—primitive oriented geometric rays paired with positive traversal multiplicities. Primitive classes give unique non-grazing, isolated, hyperbolic primitive rays. We derive the all-period primitive ledger, time reversal, length bounds, and positive determinant-one optical monodromy.

1 Model, conventions, and No-eclipse geometry

Let $c_0, c_1, c_2 \in \mathbb{R}^2$ be the vertices of an equilateral triangle of side d, and let

$$K_i = \{x \in \mathbb{R}^2 : |x - c_i| \le r\}, \qquad i = 0, 1, 2.$$

The billiard moves at unit speed in the complement of the interiors of the K_i and reflects specularly. We sample immediately after each collision, so the clock is one bounce. Rays are oriented; changing the initial collision only cyclically shifts the itinerary. Time reversal is not quotiented.

A cyclic word $w = (i_0, \dots, i_{n-1})$ is *reduced* when $i_j \ne i_{j+1}$, including $i_{n-1} \ne i_0$. It is primitive when it is not a proper power. Every cyclic class has a unique form $[w] = [u^m]$ with $[u]$ primitive and $m \ge 1$. A *periodic-ray iterate* is a primitive oriented geometric ray paired with its positive traversal multiplicity m. Length-one reduced words do not exist.

Lemma 1 (Sharp capsule gap)**.** *The convex hull of either two disks is disjoint from the third disk exactly when*

$$d > \frac{4r}{\sqrt{3}}.$$

In that chamber the separation is $\sqrt{3}d/2 - 2r > 0$.

Proof. The convex hull of two radius-r disks is the closed radius-r neighborhood of their center segment. The third center has distance $\sqrt{3}d/2$ from that segment. Removing the capsule radius and the third-disk radius leaves $\sqrt{3}d/2 - 2r$, proving both assertions. □

2 Convex variational coding

For a reduced word w, put $X_w = \prod_{j=0}^{n-1} K_{i_j}$ and, with cyclic indices, define

$$\mathcal{L}_w(q_0, \dots, q_{n-1}) = \sum_{j=0}^{n-1} |q_{j+1} - q_j|. \tag{1}$$

Theorem 1 (Complete periodic-ray coding)**.** *Assume $r > 0$ and $d > 4r/\sqrt{3}$. Cyclically reduced cyclic classes are in bijection with periodic-ray iterates. If $[w] = [u^m]$, its image is the m-fold traversal of the unique primitive oriented exterior ray coded by $[u]$. Every such geometric support is non-grazing, isolated, and hyperbolic, and*

$$[w] \longmapsto [\mathrm{rev}(w)]$$

is the time-reversal involution.

Proof. The continuous convex functional (1) attains a minimum on the compact convex set X_w. Fix a minimizing vertex q_j. If it were in the interior of K_{i_j}, then it would minimize $|x - q_{j-1}| + |q_{j+1} - x|$ without a local constraint. Equality in the triangle inequality places every such minimizer on $[q_{j-1}, q_{j+1}]$. That segment lies in the convex hull of one or two disks other than K_{i_j} and is disjoint from K_{i_j} by Lemma 1, a contradiction. Thus all vertices lie on their assigned circles.

Suppose q and q' were distinct minimizers. Convexity makes $(q + q')/2$ another minimizer. At a coordinate where they differ, strict convexity of the disk puts the midpoint in its interior, contradicting the preceding paragraph. The minimizer is therefore unique.

At a minimizing boundary vertex, the tangential part of

$$\frac{q_j - q_{j-1}}{|q_j - q_{j-1}|} - \frac{q_{j+1} - q_j}{|q_{j+1} - q_j|}$$

vanishes. The normal multiplier is nonzero: if it vanished, the two unit vectors would agree and q_j would lie between its neighbors, again contradicting no-eclipse. The incoming and outgoing tangential components therefore agree and their nonzero normal components are opposite. This is precisely non-grazing specular reflection. The supporting half-plane at an endpoint keeps a flight out of its endpoint disks, and Lemma 1 keeps it out of the remaining disk. Hence the minimizing polygon is a physical exterior ray.

Conversely, the reflection law and the exterior orientation give the normal multiplier with the minimizing sign. They are the first-order optimality conditions for the convex functional on X_w, and hence are sufficient for global minimality. Uniqueness identifies every coded iterate with the constructed minimizer. If $w = u^m$, shifting the minimizing tuple by $|u|$ coordinates preserves X_w and $\mathcal{L}_w$; uniqueness forces the shift to fix the tuple, so the polygon is the m-fold traversal of the primitive polygon for u. Conversely, a smaller geometric collision period makes w a proper power. This proves the primitive/iterate statement. For example, [01] and [0101] share one geometric support but have traversal multiplicities one and two. Reversal follows directly from the itinerary. Non-grazing makes the finite itinerary locally constant, while uniqueness rules out a neighboring periodic ray with that itinerary; equivalently, the hyperbolicity proved below also gives isolation. □

Why the proof is not a finite computation. Compactness gives existence for an arbitrary word length. No-eclipse and strict convexity, not an enumeration cutoff, give boundary contact and uniqueness. The converse is included because a stationary reflected polygon would otherwise not automatically be the unique convex minimizer.

3 Monodromy and uniform length control

Let $\ell_j > 0$ be a free-flight length and let ϕ_j be the incidence angle measured from the obstacle normal, so $\cos\phi_j > 0$. In transverse Jacobi coordinates, free flight followed by a dispersing reflection is represented, up to the standard collision-chart conjugacy, by

$$B_j = R(a_j)F(\ell_j) = \begin{pmatrix} 1 & 0 \\ a_j & 1 \end{pmatrix} \begin{pmatrix} 1 & \ell_j \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & \ell_j \\ a_j & 1 + a_j\ell_j \end{pmatrix}, \qquad a_j = \frac{2}{r\cos\phi_j} > 0. \tag{2}$$

Proposition 1 (Dispersing hyperbolicity)**.** *For every periodic ray, the monodromy $M = B_{n-1}\cdots B_0$ has determinant one, positive entries, and $\operatorname{tr} M > 2$. Thus its multipliers are real positive reciprocals Λ, Λ^{-1} with $\Lambda > 1$.*

Proof. Each factor, and hence their product, has determinant one and strictly positive entries. Thus $M_{11}M_{22} = 1 + M_{12}M_{21} > 1$, and the arithmetic–geometric mean inequality gives $\operatorname{tr} M = M_{11} + M_{22} \geq 2\sqrt{M_{11}M_{22}} > 2$. The characteristic polynomial $\lambda^2 - (\operatorname{tr} M)\lambda + 1$ has two distinct positive reciprocal roots, one larger than one. □

Every flight joins two distinct disks whose centers are distance d apart, so the triangle inequalities give

$$d - 2r \leq \ell_j \leq d + 2r, \qquad n(d - 2r) \leq L_w \leq n(d + 2r). \tag{3}$$

At $r = 1, d = 3$, the symmetric orbit [01] has two flights of length one and $a = 2$. Its block squared is

$$\begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}^2 = \begin{pmatrix} 3 & 4 \\ 8 & 11 \end{pmatrix}, \qquad \operatorname{tr} M = 14.$$

The symmetric orbit [012] has flight length $3 - \sqrt{3}$, incidence cosine $\sqrt{3}/2$, and total length $9 - 3\sqrt{3}$; its positive optical product is again hyperbolic.

4 Exact symbolic and primitive ledgers

Let $A = J_3 - I_3$ be the adjacency matrix: $A_{ij} = 1$ exactly when $i \neq j$. Its eigenvalues are $2, -1, -1$. Therefore the number of collision-marked reduced n-bounce return records, including iterates whose primitive periods divide n, is

$$F_n = \operatorname{tr}(A^n) = 2^n + 2(-1)^n. \tag{4}$$

Every exact-period rooted word has a free orbit of size n under cyclic shift. Möbius inversion consequently gives

$$P_n = \sum_{e|n} \mu(e) F_{n/e}, \qquad O_n = \frac{P_n}{n}, \tag{5}$$

where O_n counts primitive oriented geometric rays of collision period n. Reversal can fix some cyclic classes, so it is an involution rather than permission to divide all O_n by two.

With bounce count as the only variable, the source collision-code zeta is

$$\zeta_{\mathrm{coll}}(z) = \exp\left(\sum_{n\geq 1} \frac{F_n}{n} z^n\right) = \det(I - zA)^{-1} = \frac{1}{(1-2z)(1+z)^2}. \tag{6}$$

Equations (4)–(6) are exact for every period because Theorem 1 supplies the geometric bridge.

Source lineage

Sinai's dispersing-billiard work supplies historical context [1]; Ikawa supplies neighboring several-convex-body and no-eclipse context [2]; Gaspard and Rice are a direct owner in the three-hard-disk scattering neighborhood [3]. These citations do not assert novelty priority, and the proof above is self-contained.

References

[1] Y. G. Sinai, "Dynamical systems with elastic reflections: ergodic properties of dispersing billiards," *Russian Mathematical Surveys* 25 (1970). DOI: 10.1070/RM1970v025n02ABEH003794.

[2] M. Ikawa, "Decay of solutions of the wave equation in the exterior of several convex bodies," *Annales de l'Institut Fourier* 38 (1988). DOI: 10.5802/aif.1137.

[3] P. Gaspard and S. A. Rice, "Semiclassical quantization of the scattering from a classically chaotic repellor," *Journal of Chemical Physics* 90 (1989). DOI: 10.1063/1.456019.